Results: subject-wise 4-fold CV (1-s windows, 75 % overlap)

Mean ± std over folds. Positive class = more severe group. Subject-level prediction = mean of the subject's 1,197 segment probabilities.

Subject-level

Task	Model	ACC	Bal. ACC	F1	Precision	Recall	Specificity	AUC
None vs (Mild + Severe)	EEGNet	0.605 ± 0.113	0.591 ± 0.098	0.501 ± 0.066	0.657 ± 0.262	0.438 ± 0.075	0.743 ± 0.248	0.569 ± 0.128
None vs (Mild + Severe)	ShallowFBCSPNet	0.524 ± 0.072	0.513 ± 0.054	0.405 ± 0.085	0.485 ± 0.085	0.385 ± 0.159	0.642 ± 0.234	0.567 ± 0.113
None vs (Mild + Severe)	Deep4Net	0.540 ± 0.089	0.525 ± 0.085	0.406 ± 0.115	0.500 ± 0.138	0.365 ± 0.137	0.684 ± 0.172	0.564 ± 0.116
(None + Mild) vs Severe	EEGNet	0.605 ± 0.103	0.441 ± 0.061	0.145 ± 0.027	0.122 ± 0.039	0.192 ± 0.017	0.690 ± 0.131	0.409 ± 0.133
(None + Mild) vs Severe	ShallowFBCSPNet	0.556 ± 0.103	0.459 ± 0.139	0.198 ± 0.150	0.145 ± 0.111	0.317 ± 0.233	0.602 ± 0.104	0.387 ± 0.169
(None + Mild) vs Severe	Deep4Net	0.532 ± 0.180	0.468 ± 0.052	0.168 ± 0.132	0.110 ± 0.084	0.375 ± 0.350	0.562 ± 0.279	0.409 ± 0.171
Mild vs Severe	EEGNet	0.563 ± 0.061	0.584 ± 0.093	0.507 ± 0.114	0.451 ± 0.061	0.633 ± 0.275	0.535 ± 0.146	0.638 ± 0.175
Mild vs Severe	ShallowFBCSPNet	0.599 ± 0.078	0.591 ± 0.103	0.469 ± 0.177	0.484 ± 0.184	0.525 ± 0.340	0.656 ± 0.232	0.555 ± 0.087
Mild vs Severe	Deep4Net	0.508 ± 0.094	0.543 ± 0.106	0.497 ± 0.166	0.399 ± 0.100	0.700 ± 0.346	0.385 ± 0.198	0.550 ± 0.149

Comparison with non-overlapping windows

Subject-level means over the 4 folds. Same folds, seeds and training settings; only the window step differs (1 s vs 0.25 s).

Task	Model	ACC: no overlap	ACC: 75 % overlap	Bal. ACC: no overlap	Bal. ACC: 75 % overlap	AUC: no overlap	AUC: 75 % overlap
None vs (Mild + Severe)	EEGNet	0.556	0.605	0.532	0.591	0.575	0.569
None vs (Mild + Severe)	ShallowFBCSPNet	0.589	0.524	0.584	0.513	0.563	0.567
None vs (Mild + Severe)	Deep4Net	0.597	0.540	0.579	0.525	0.560	0.564
(None + Mild) vs Severe	EEGNet	0.669	0.605	0.466	0.441	0.453	0.409
(None + Mild) vs Severe	ShallowFBCSPNet	0.597	0.556	0.447	0.459	0.427	0.387
(None + Mild) vs Severe	Deep4Net	0.637	0.532	0.445	0.468	0.428	0.409
Mild vs Severe	EEGNet	0.453	0.563	0.500	0.584	0.579	0.638
Mild vs Severe	ShallowFBCSPNet	0.618	0.599	0.592	0.591	0.637	0.555
Mild vs Severe	Deep4Net	0.560	0.508	0.584	0.543	0.573	0.550

Experimental settings

All three models use the same settings and the same folds. No hyper-parameters were tuned; the values below were fixed before the run.

Category	Setting	Value
Input	Channels / window	F3, F4 / middle 300 s of each recording
	Sampling rate	128 Hz (from 1000 Hz, polyphase resampling)
	Segments	1 s windows, **75 % overlap** (0.25-s step), 1,197 per subject (148,428 in total)
	Normalisation	per-channel z-score, mean/std from the training subjects of each fold
Cross-validation	Scheme	stratified 4-fold, **split by subject** (all segments of a subject are in one fold)
	Validation set	20 % of the training subjects (stratified), used only for early stopping
	Subjects per fold (train / val / test)	Task 1–2: 74 / 19 / 31; Task 3: 32–33 / 9 / 13–14
	Random seed	42 (fold split); 42 + fold index (inner split, weight init, batch order)
	Repetitions	1 (a single 4-fold CV run)
Training	Loss	cross-entropy weighted by inverse class frequency (training set)
	Optimiser	AdamW, lr 1e-3, weight decay 1e-4
	LR schedule	cosine annealing over 50 epochs
	Batch size	128 segments (≈ 4× more steps per epoch than without overlap)
	Epochs	max 50; early stopping on validation loss, patience 10; best-validation weights restored
Evaluation	Segment level	each 1-s segment classified independently
	Subject level	mean of the subject's 1,197 segment probabilities, threshold 0.5
	Metrics	ACC, balanced ACC, precision, recall (sensitivity), specificity, F1, macro-F1, ROC-AUC; positive class = more severe group; mean ± std over the 4 test folds
	Baseline	majority-class ACC (always predicting the larger group)
Complexity	Params / size	all parameters and buffers, fp32
	FLOPs	`torch.utils.flop_counter.FlopCounterMode` for one 2 × 128 input; conv + linear layers, 1 multiply-add = 2 FLOPs

Subjects are split into folds. Splitting by segment would put 1-s pieces of the same recording in both training and test sets. The model could then recognise the person instead of their depression level, which inflates accuracy. With 75 % overlap this would be even worse, because neighbouring windows share 96 of their 128 samples. The overlap is applied to training, validation and test windows; since every subject is in exactly one fold, overlapping windows never cross the train/test boundary.

Models

Model	Configuration	Params	Size (fp32)	FLOPs / 1-s segment
EEGNet (`EEGNetv4`)	braindecode defaults (EEGNet-8,2; kernel 64 = 0.5 s at 128 Hz)	1,266	5.3 KB	0.31 M
ShallowFBCSPNet	braindecode defaults	4,482	17.8 KB	0.42 M
Deep4Net	kernels 5, pools 2/2 (see below)	135,002	530 KB	3.64 M

The default Deep4Net (temporal kernel 10, max-pool 3/3 in four blocks, designed for 250 Hz / 4-s trials) cannot process a 128-sample input: the time axis shrinks below 1. Kernels and pools are scaled to roughly 128/250 of the original (kernel 5, pool 2 stride 2), giving time lengths 128 → 62 → 29 → 12 → 4. Filter counts, dropout and everything else are unchanged.